

National Computer Education Accreditation Council - NCEAC

NCEAC.FORM.001C

INSTITUTION: National University of Computer & Emerging Sciences, Karachi.

Computer Science - Fall 2024

PROGRAM (S) TO BE EVALUATED

Course Description

Course Code	CL1000																							
Course Title	Introduction to Information and Communication Technology (IICT)																							
Credit Hours	1 (3 Lab contact hours)																							
Prerequisites by Course(s) and Topics	None																							
Grading Policy	Absolute as per NUCES grading policy																							
Policy about missed assessment items in the course	Retake of missed assessment items (other than midterm/ final exam) will not be held. For a missed mid-term/ final exam, an exam retake/ pre-take application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decide the exam retake/ pre-take cases.																							
Course Plagiarism Policy	- Plagiarism in project or midterm/ final exam may result in F grade in the course. - Plagiarism in an assignment will result in zero marks in the whole assignments category.																							
URL (if any)																								
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	1 credit, 3 hours contact labs Breakdown Assessment Items <table><tr><th>Assessment Item</th><th>Number</th><th>Weight (%)</th></tr><tr><td>Assignments/HWs</td><td>2</td><td>5</td></tr><tr><td>Quizzes</td><td>2</td><td>5</td></tr><tr><td>Project</td><td>1</td><td>5</td></tr><tr><td>Lab Work</td><td>Best 10 labs</td><td>10</td></tr><tr><td>Mid Exam</td><td>1</td><td>25</td></tr><tr><td>Final Exam</td><td>1</td><td>50</td></tr></table>			Assessment Item	Number	Weight (%)	Assignments/HWs	2	5	Quizzes	2	5	Project	1	5	Lab Work	Best 10 labs	10	Mid Exam	1	25	Final Exam	1	50
Assessment Item	Number	Weight (%)																						
Assignments/HWs	2	5																						
Quizzes	2	5																						
Project	1	5																						
Lab Work	Best 10 labs	10																						
Mid Exam	1	25																						
Final Exam	1	50																						
Course Committee	Prof. Dr. Atif Tahir (HOS), Dr. Muhammad Rafi (HOD AI & DS), Dr. Abdul Aziz (HOD SE), Dr. Fahad Samad (HOD CY)																							
Lab Coordinator	Mr. Talha Shahid																							
Current Catalog Description	History of Computer Systems, Hardware, Software and Architectures. Introduction to Information Systems. Basic Number Systems Arithmetic. Learning of Applications Software such as Document Processing, Spreadsheet, Presentation and Database. System Software Development Web and Internet Services.																							

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	Introduction to Operating System, Computer Networks and related emerging technologies.		
Textbook (or Laboratory Manual for Laboratory Courses)	1. Charles S. Parker, Understanding Computers: Today and Tomorrow, Course Technology, 25 Thomson Place, Boston, Massachusetts 02210, USA 2. Livesley, Robert Kenneth. An introduction to automatic digital computers. Cambridge University Press, 2017.		
Reference Material	3. Zawacki-Richter, Olaf, and Colin Latchem. "Exploring four decades of research in Computers & Education." Computers & Education 122 (2018): 136-152. 4. Sinha, Pradeep K., and Priti Sinha. Computer fundamentals. BPB publications, 2010. - Goel, Anita. Computer fundamentals. Pearson Education India, 2010.		
Course Detailed Description & CLOs	Brief history of Computer, Four Stages of History, Computer Elements, Processor, Memory, Hardware, Software, Application Software its uses and Limitations, System Software its Importance and its Types, Types of Computers (Super, Mainframe, Mini and Micro Computer), Introduction to CBIS (Computer Based Information System), Methods of Input and Processing, Class2. Organizing Computer Facility, Centralized Computing Facility, Distributed Computing Facility, and Decentralized Computing Facility, Input Devices. Keyboard and its Types, Terminal (Dumb, Smart, Intelligent), Dedicated Data Entry, SDA (Source Data Automation), Pointing Devices, Voice Input, Output Devices. Soft- Hard Copies, Monitors and its Types, Printers and its Types, Plotters, Computer Virus and its Forms, Storage Units, Primary and Secondary Memories, RAM and its Types, Cache, Hard Disks, Working of Hard Disk, Diskettes, RAID, Optical Disk Storages (DVD, CD ROM), Magnetic Types, Backup System, Data Communications, Data Communication Model, Data Transmission, Digital and Analog Transmission, Modems, Asynchronous and Synchronous Transmission, Simplex, Half Duplex, Full Duplex Transmission, Communications, Medias (Cables, Wireless), Protocols, Network Topologies (Star, Bus, Ring), LAN, LAN, Internet, A Brief History, Birthplace of ARPA Net, Web Link, Browser, Internet Services provider and Online Services Providers, Function and Features of Browser, Search Engines, Some Common Services available on Internet. The Course Learning Objective (CLOs) is to prepare students to become effective users of ICT so as to support their life-long learning, as well as to provide a pathway into the workforce. After completion of the course, the student shall be able to: 1. Learn and understanding of Information, Computer Systems Software and Hardware Technologies. 2. Learning of basic logic, algorithms and number system arithmetic. 3. Understanding and practical knowledge of Applications Software. 4. Learning and experience of basic Web Software Development with internet. 5. Knowledge of Operating System and Network Communication Technologies.		
	PLO 1	Computing Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing ✓

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			specialization to the solution of complex computing problems.	
	PLO 2	Problem Analysis	Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.	✓
	PLO 3	Design/ Develop Solutions	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	
	PLO 4	Investigation & Experimentation	Conduct investigation of complex computing problems using research-based knowledge and research-based methods	✓
	PLO 5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.	✓
	PLO 6	Society Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.	
	PLO 7	Environment and Sustainability	Understand and evaluate sustainability and impact of professional computing work in the solution of complex computing problems	
	PLO 8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.	
	PLO 9	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.	
	PLO 10	Communication	Communicate effectively on complex computing activities with the computing community and with society at large.	✓
	PLO 11	Project Management and Finance	Demonstrate knowledge and understanding of management principles and economic decision	

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			making and apply these to one's own work as a member or a team.											
	PLO 12	Life Long Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.	✓										
C. Mapping of CLOs on PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)														
		PLOs												
			1	2	3	4	5	6	7	8	9	10	11	12
CLOs	1	✓	✓									✓		
	2	✓	✓		✓									
	3	✓			✓	✓						✓		✓
	4	✓	✓		✓	✓						✓		✓
	5	✓				✓						✓		✓

Topics Covered in the Course/Lab, with Number of Labs on Each Topic	Topics to be covered in the lab:			
	List of Topics	No. of Weeks	Contact Hours	CLOs
	Introduction and brief history of computer, information systems and its applications.	2	6	1
	Computer Hardware & Architecture.	1	3	1
	Number Systems and their conversions.	1	3	2
	Introduction to Operating Systems, Files & Control Panel.	1	3	3
	Preparing Professional Documents using Microsoft Word.	2	6	3
	Format & Organize Data using Microsoft Excel	2	6	3
	Preparing Professional Presentations using MS Power Point.	1	3	3
	Developing Databases Forms & Reports using MS Access.	1	3	4
	Introduction to Communications & Computer Networks.	1	3	5
	Basic Web Template Development using HTML, CSS.	1	3	4
	JavaScript and Web development.	1	3	4
Introduction to latex and its basics.	1	3	2	

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	Object oriented programming basic theory.	1	2	3
	Total	16	48	
Laboratory Projects/Experiments Done in the Course	Hands on practical experience of MS Windows, MS Word, MS Excel, MS Access, MS Power Point, WordPress.			
Programming Assignments Done in the Course	Preparing professional documents and Presentation Slides, Basic Data Management, Quick Web Development, Basic Operating System & Network Troubleshooting, Object oriented programming.			
Class Time Spent on (in credit hours, Hrs./Min)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	1.5/90	1.2/70	1.2/70	0.1/10
Oral and Written Communications	Demo, Presentations, Group Discussions			

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Lab Contents				
Weeks	Contents/Topics	Labs/ Manuals	Deliverable	Comments (if any)
Week-01	Introduction to Computers and Technology: Processing Life Cycle, Types, and Components	Features of MS Word and MS Excel		
Week-02	Memory (RAM/ROM), Storage Devices, and Operating Systems: Types, File Management, and Restoration	Features of MS PowerPoint and Essential Command Prompt Commands for Windows		
Week-03	Arithmetic and Conversions in Number Systems.	Paper-Based Conversion Tasks and Techniques		
Week-04	Introduction to Databases/DBMS: Creating Tables, Key Columns, Data Insertion, and Relationships	MS Access Features and SQL Queries 1	Quiz 01	
Week-05	Basic SQL, Forms, and Reports.	MS Access Features and SQL Queries 2	Assignment 01	
Week-06	Theory Mid-Exam 01			
Week-07	HTML, CSS, bootstrap library, commands and working	Lab to develop simple web-pages		
Week-08	Lab Mid-Exam			

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Week-09	HTML, CSS, JavaScript, Functions and libraries.	Interactive web-development	Quiz 02	
Week-10	Introduction to Communications & Computer Networks: Network Topologies, Components, Protocols, IP Addresses, and Connections.	Packet tracer intro, basic network connectivity.	Assignment 02	
Week-11	Introduction to latex and its basics.	Overleaf		
Week -12	Theory Mid-Exam 02			
Week 13	Object oriented programming basic theory.	Classes and Objects.		
Week-14	Prompt Engineering Basic.	Basic Prompt Engineering: Crafting and Refining Effective Prompts		
Week-15	Project Evaluation and Presentation.			
Week-16	Lab Final-Exam			

Instructor Name:

Instructor Signature:

Date: